PROJECT PALEOWAVE

Ichthyosaur Fossil Locality Intelligence Report

Humboldt Range & Gabbs Valley — Central Nevada

Generated: March 2026 | Model Version: RF-v1 | AUC: 0.906

MISSION: Identify high-probability ichthyosaur fossil localities in central Nevada using machine learning on USGS terrain data, cross-validated against PBDB occurrence records and Nevada Triassic marine geology. This report provides actionable field targets ranked by composite probability score.

■ IMPORTANT DISCLAIMERS & SAFETY INFORMATION

RESEARCH TOOL — NOT A GUARANTEE

This report is generated by a machine learning model trained on publicly available terrain and paleontological data. Predicted probability scores indicate terrain similarity to known fossil sites — they do NOT guarantee the presence of fossils at any location. Many high-scoring sites will yield no fossils. Field results will vary significantly.

KNOW THE LAW BEFORE YOU GO

Fossil collection laws in the United States are complex and vary by land ownership. On BLM and U.S. Forest Service land, casual collection of reasonable quantities of common invertebrate and plant fossils for personal use is generally permitted without a permit. HOWEVER: Collection of vertebrate fossils (including ichthyosaurs and all other reptile, fish, and mammal remains) on federal land requires a Paleontological Resources Preservation Act (PRPA) permit. Unpermitted collection of vertebrate fossils on federal land is a federal crime punishable by fines up to $20,000 and/or imprisonment. Contact the BLM Winnemucca District Office (775-623-1500) BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates in this report are based on geographic analysis only. Land ownership has NOT been verified for individual sites. Sites may fall on BLM, U.S. Forest Service, tribal, state, private, or other land with different regulations. Always verify land ownership using the BLM's GeoCommunicator or Nevada state GIS resources before entering any property. Trespassing on private or tribal land is illegal. Obtain written permission where required.

REMOTE TERRAIN & PERSONAL SAFETY

Target sites in the Humboldt Range and Gabbs Valley area are in extremely remote, rugged terrain with no cell service, no water sources, extreme temperature swings, and significant flash flood risk in canyon areas. Do not venture into these areas alone. Always carry a satellite communicator (e.g. Garmin inReach), sufficient water (minimum 4L/person/day in summer), a first aid kit, and a paper topo map. Inform someone of your planned route and expected return before departure. The authors of this report accept no liability for injury, death, or legal consequences arising from use of this document.

SITE DISTURBANCE & SCIENTIFIC INTEGRITY

Even where collection is legal, unnecessary disturbance of fossil sites degrades their scientific value. Photograph everything in place before moving it. Record precise GPS coordinates for any finds. Report significant discoveries to the Nevada State Museum (nsm.nevadaculture.org) and consider contributing locality data to the Paleobiology Database (paleobiodb.org) to improve future research.

MODEL LIMITATIONS

This model uses terrain data only. It does not account for subsurface geology, fault displacement, erosion history, formation thickness, or prior collection effort. The training dataset contains only 27 positive examples — a small sample that may not capture the full range of ichthyosaur-bearing terrain. Model predictions should be treated as one input among many, not as definitive prospecting guidance. Consult a professional geologist or paleontologist before making significant decisions based on this output.

By using this report you acknowledge these disclaimers and accept full personal responsibility for compliance with all applicable laws, regulations, and safety precautions.

MODEL SUMMARY

Metric	Value	Notes
Training positives	27	PBDB ichthyosaur occurrences, Nevada
Background points	270	Random terrain samples, 10:1 ratio
Algorithm	Random Forest	500 trees, max depth 6, balanced classes
Cross-validation AUC	0.906 +/- 0.072	5-fold stratified CV
Top feature	Slope (0.293)	Closely followed by TRI (0.291)
DEM source	USGS 3DEP 1/3 arc-second	~10m resolution, 6 tiles merged
Study area	700M+ pixels	n39-41, w117-119 coverage
Candidate sites	50	Top-ranked from full prediction surface
Geology source	USGS NGMDB Nevada 1:500k	Triassic marine units filtered

FEATURE IMPORTANCE

Terrain metrics ranked by Random Forest importance. Slope and ruggedness together explain 58% of predictive power, confirming ichthyosaur fossils in Nevada preferentially occur on moderately inclined, eroding ridge flanks.

Feature	Importance	Interpretation
Slope (degrees)	**0.293**	Moderate slopes 8-15 degrees favor exposed outcrops
Ruggedness (TRI)	**0.291**	Low-moderate roughness — eroding but accessible
Aspect (degrees)	**0.222**	Both N and S facing slopes represented
Elevation (m)	**0.195**	Sweet spot 1400-1750m, mean 1647m

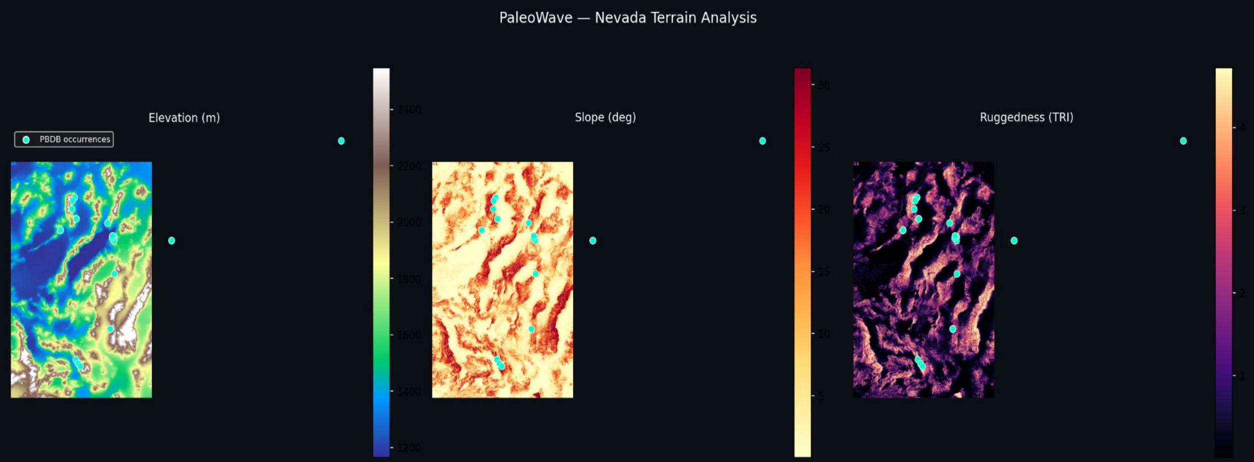

Figure 1: Nevada terrain analysis — elevation, slope, and ruggedness with PBDB occurrences (teal dots). Fossil sites cluster on mid-elevation ridge flanks with moderate slope.

TOP 10 PRIORITY FIELD TARGETS

Ranked by composite score: ML terrain probability + geology bonus for proximity to TRc (Triassic limestone/dolomite — includes Favret, Luning, Star Peak Group, Augusta Mountain formations). See disclaimer page regarding permits required before collecting vertebrate fossils.

#1	**40.404722N, -118.243889W** 40° 24' 17.0" N 118° 14' 38.0" W	**ML Probability:** 0.8005 **Geology:** ON TRc formation (52m) **Formation:** TRc — Triassic limestone/dolomite	1.0000
#2	**40.209167N, -117.587778W** 40° 12' 33.0" N 117° 35' 16.0" W	**ML Probability:** 0.7952 **Geology:** 992m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8952
#3	**40.550278N, -118.232500W** 40° 33' 1.0" N 118° 13' 57.0" W	**ML Probability:** 0.7946 **Geology:** 1361m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8946
#4	**40.434444N, -117.686667W** 40° 26' 4.0" N 117° 41' 12.0" W	**ML Probability:** 0.8423 **Geology:** 4781m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8923
#5	**40.839722N, -117.720833W** 40° 50' 23.0" N 117° 43' 15.0" W	**ML Probability:** 0.7920 **Geology:** 1059m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8920
#6	**40.822778N, -117.696944W** 40° 49' 22.0" N 117° 41' 49.0" W	**ML Probability:** 0.7899 **Geology:** 1309m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8899
#7	**40.418889N, -117.704444W** 40° 25' 8.0" N 117° 42' 16.0" W	**ML Probability:** 0.8305 **Geology:** 3338m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8805
#8	**40.787222N, -117.456389W** 40° 47' 14.0" N 117° 27' 23.0" W	**ML Probability:** 0.8533 **Geology:** 13040m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8533
#9	**40.907222N, -118.464444W** 40° 54' 26.0" N 118° 27' 52.0" W	**ML Probability:** 0.8523 **Geology:** 41459m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8523
#10	**40.113056N, -117.202500W** 40° 6' 47.0" N 117° 12' 9.0" W	**ML Probability:** 0.7991 **Geology:** 4801m to TRc formation **Formation:** TRc — Triassic limestone/dolomite	0.8491

PREDICTION SURFACE & HOTSPOT MAP

Left: Full fossil probability surface. Right: Top 5% hotspots. White dots = known PBDB occurrences. Gold triangles = ML candidate sites.

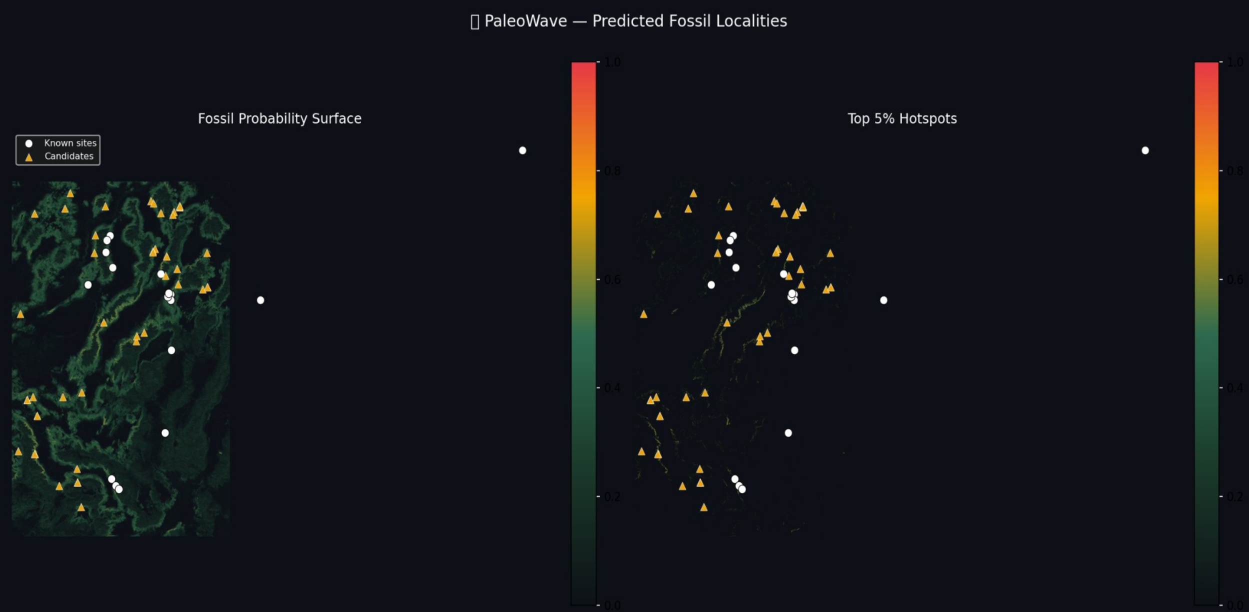

Figure 2: Fossil probability surface and top 5% hotspots.

PRIMARY TARGET ZONE — HUMBOLDT RANGE

Priority #1 site (40.4047N, 118.2439W) sits 52m from confirmed TRc Triassic marine limestone. Purple/white = ML probability surface. Cyan = Triassic marine formations. Colored dots = priority targets 1-50.

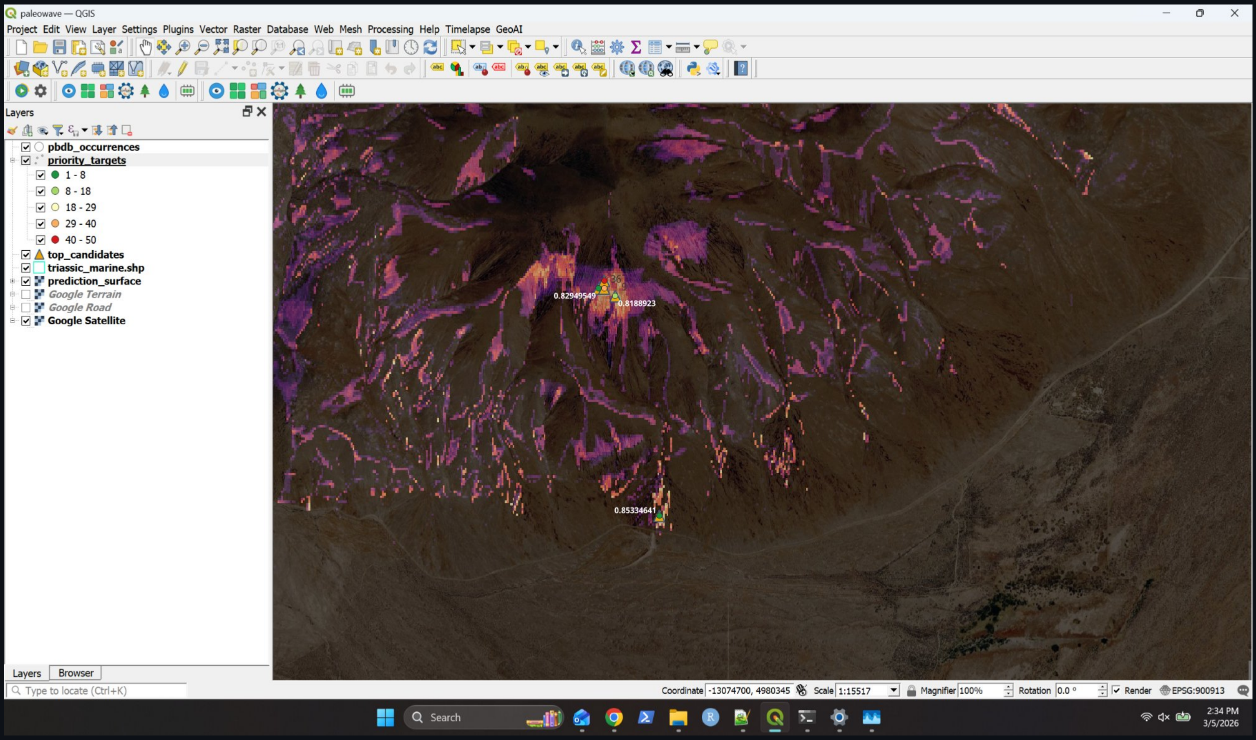

Figure 3: QGIS view — primary Humboldt Range target cluster.

FIELD PREPARATION NOTES

Access

The Humboldt Range target cluster is accessed via US-95 north from Lovelock (NV), then east on dirt roads toward the range front. High-clearance 4WD required. Check BLM road conditions before departure. Verify land status before entering.

Best Season

April-June and September-October. Avoid July-August heat (38C+) and January-February snow. Spring provides best erosional surface exposure.

What to Look For

Triassic limestone and dolomite outcrops (grey to buff, often platy or nodular). Dark grey chert nodules within limestone — characteristic of Favret and Prida formations. Bone material appears as darker, denser inclusions with subtle texture difference from matrix.

Permits — READ THIS

Vertebrate fossil collection on federal land requires a PRPA permit. Contact BLM Winnemucca District Office (775-623-1500) BEFORE collecting any vertebrate material. Unpermitted collection is a federal crime.

Safety Equipment

GPS with loaded coordinates, rock hammer, hand lens (10x min), sample bags, field notebook, camera, topo maps, satellite communicator (no cell service), minimum 4L water per person per day, first aid kit.

Reporting Finds

Report significant finds to Nevada State Museum and contribute locality data to the Paleobiology Database (paleobiodb.org) to improve future research.

QUICK REFERENCE — TARGET COORDINATES

Load priority_targets.geojson into GPS or mapping app for all 50 sites.

#	Decimal Degrees	DMS Latitude	DMS Longitude	Score
1	40.404722, -118.243889	40° 24' 17.0" N	118° 14' 38.0" W	1.0000
2	40.209167, -117.587778	40° 12' 33.0" N	117° 35' 16.0" W	0.8952
3	40.550278, -118.232500	40° 33' 1.0" N	118° 13' 57.0" W	0.8946
4	40.434444, -117.686667	40° 26' 4.0" N	117° 41' 12.0" W	0.8923
5	40.839722, -117.720833	40° 50' 23.0" N	117° 43' 15.0" W	0.8920
6	40.822778, -117.696944	40° 49' 22.0" N	117° 41' 49.0" W	0.8899
7	40.418889, -117.704444	40° 25' 8.0" N	117° 42' 16.0" W	0.8805
8	40.787222, -117.456389	40° 47' 14.0" N	117° 27' 23.0" W	0.8533
9	40.907222, -118.464444	40° 54' 26.0" N	118° 27' 52.0" W	0.8523
10	40.113056, -117.202500	40° 6' 47.0" N	117° 12' 9.0" W	0.8491